Late Breaking Results: BLAST: Bisection-Free Learning Approach for Statistical Timing Characterization

Kai Jing, Tao Bai, Zeyuan Deng, Junming Jiao, and Peng Cao*
National ASIC System Engineering Technology Research Center, Southeast University, Nanjing, China
caopeng@seu.edu.cn

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***Abstract*—Statistical timing characterization for standard cells faces significant computational challenges due to the laborious bisection analysis for setup/hold constraint of sequential cells. To address this issue, we propose a Bisection-Free Learning Approach for Statistical Timing Characterization (BLAST) by extracting inherent delay for data path and clock path in sequential cells as specific features. Multi-task learning is implemented with a multi-gate mixture-of-experts (MMoE) model to exploit the profound interdependency between setup and hold constraint for different timing arcs, where the active learning strategy is incorporated to improve learning efficiency. Experimental results under 135 PVT corners with TSMC 12nm process demonstrate that the proposed BLAST achieves considerable acceleration by avoiding the iterative bisection search for statistical constraint prediction with 76.9% runtime reduction compared to the commercial tool. Excellent prediction accuracy is achieved for various flip-flops by BLAST with the relative root mean square error (rRMSE) of 2.21% and worst-case absolute error (WCAE) of 0.82 *ps*.**

***Index Terms*—Statistical characterization, setup/hold constraint, multi-task learning, active learning**

I. Introduction

Standard cell libraries are crucial for integrated circuit design, but their characterization process is performed by time-consuming transistor-level simulations under various process, voltage, and temperature (PVT) corners, especially for statistical characterization in modern technologies. Different from combinational cells. the timing constraints for sequential cells need to be characterized by bisection methodology, which require tens of iterations of binary search for each constraint [1], further exacerbating simulation costs.

Considerable efforts have been dedicated to standard cell characterization to achieve tradeoff between accuracy and simulation overhead. Machine learning techniques have been utilized to enhance characterization efficiency by exploiting correlations between timing characteristics and physical parameters [2, 3]. However, these methods often overlook the complex internal relationships among timing arcs. For sequential cell characterization, delta delay serves as an approximation of timing constraints to significantly reduce simulation costs from binary search. [4]. However, as advanced processes evolve, the mismatch between the delta delay and traditional bisection methodology has become increasingly pronounced.

In order to achieve accurate and efficient statistical timing constraint characterization for sequential cells, we propose a Bisection-Free Learning Approach for Statistical Timing Characterization (BLAST), which eliminates the labor-intensive iteration costs inherent in traditional bisection methods. The multi-gate mixture-of-experts (MMoE) model is employed to capture the correlations across multiple constraint timing arcs by extracting the delta delays of setup and hold constraint as features while the active learning is utilized to select the most informative training data to improve efficiency.

II. Proposed Approach

A. Overview

The overview of proposed BLAST is illustrated in Fig. 1. During the process of feature extraction (Fig. 1(a)), the shared features and

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*Peng Cao is the corresponding author.

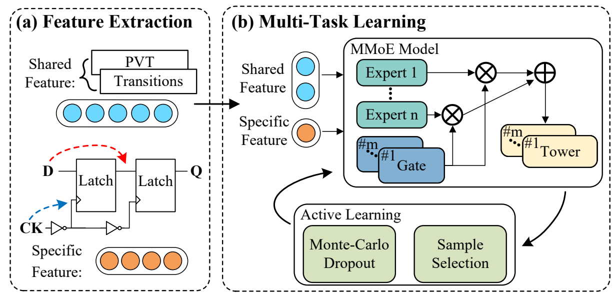

Fig. 1 Overview of BLAST.

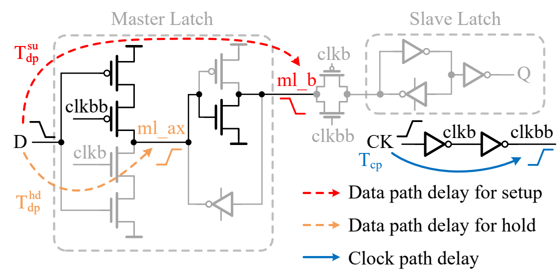

Fig. 2 Extraction procedure of specific feature.

task-specific features are extracted for the statistical setup and hold constraints prediction, where the shared features are associated with the global information for all rise and fall timing arcs of setup and hold constraints while the task-specific features are related to the timing characteristics during each specific working condition of sequential cell. During the multi-task learning process(Fig. 1(b)), the statistical setup and hold constraint characterizations for rise and fall arcs are considered as multiple tasks to be predicted by the employed MMoE model. Based on the uncertainty from the MMoE model's predictions, active learning is incorporated to refine the training process by selecting the most informative samples, thereby accelerating model convergence and enhancing prediction accuracy.

B. Feature Extraction

In accordance with traditional characterization method, the statistical setup and hold constraints for both rise and fall timing arcs are closely associated with the common factors of the sequential cell including PVT information as well as clock and data transitions, which are extracted as shared features for model learning. Moreover, it could be noticed that, in addition to the success/failure status verified by traditional bisection analysis, the setup and hold constraints are notably manifested as the competitive relationship between internal nodes of sequential cell. Inspired by this, the specific feature for each statistical timing constraint is extracted as the difference between switching delays at the appropriate internal nodes.

The extraction procedure for the specific feature is illustrated in Fig. 2. By taking a typical master-slave flip-flop (MSFF) as an example, to satisfy the setup constraint, the internal node *ml_b* after master latch must stabilize before the internal clock node *clkbb* switches, allowing effective data to propagate to the output. Considering this, the setup time constraint is closely linked to two key switching delays including the **data path delay** for setup constraint

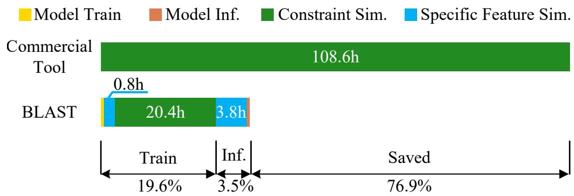

Fig. 3 Runtime comparison between commercial and proposed BLAST.

(T_{dp}^{su}) and the **clock path delay** (T_{cp}), where T_{dp}^{su} is the delay from the input data *D* to the stabilization of the internal node *ml_b* while T_{cp} is the delay from the clock signal *CK* to the internal clock node *clkbb*. Consequently, the setup constraint can be approximated as the difference between these two switching delays. Similarly, for the hold constraint, the input data signal *D* must not propagate to the internal node *ml_ax* before the internal clock node *clkbb* switches, which indicates that the hold constraint is also closely related to the **data path delay** for hold constraint (T_{dp}^{hd}) and the **clock path delay** (T_{cp}) so that can be approximated as the difference between them. For statistical characterization, we can measure these switching delays through transient simulations with Monte Carlo method and compute their standard deviations to approximate the statistical constraint $\sigma\left(\Delta T\right)$ by root sum square (RSS) approach, as given in (1).

$$\sigma\left(\Delta T\right) = \sqrt{\sigma^2\left(T_{dp}\right) + \sigma^2\left(T_{cp}\right)} \tag{1}$$

It is worth noting that although illustrated with MSFF, the extracted specific feature is applicable to various structures of sequential cells due to the consistently strong correlation between the timing constraint and the inherent propagation delay of data and clock paths.

C. *Multi-Task Learning*

Due to the profound relationship among the setup and hold constraints for different timing arcs, multi-task learning is employed to exploit the inherent interdependency by utilizing the MMoE model. As illustrated in Fig. 1(b), the MMoE model is implemented to predict the setup and hold constraints of different timing arcs simultaneously with both shared and specific features, which consists of expert, gate, and tower networks. The expert networks are responsible for learning shared features, while the gate networks assign weights to the outputs of the expert networks based on the input data. The tower networks then generate task-specific outputs, enabling the model to effectively capture the interdependencies between setup and hold constraints. This architecture ensures that both shared and specific features are utilized to improve prediction accuracy, enabling the model to make more precise predictions.

To further optimize the training efficiency, the active learning strategy is incorporated in Fig. 1(b), which selectively prioritizes the most informative samples for training. Given the high computational cost of Monte Carlo simulations for statistical constraint characterization, active learning addresses this by evaluating the uncertainty in the model's predictions and selecting uncertain samples for further simulation. By using Monte Carlo dropout as a query strategy, the model introduces variability by randomly deactivating neurons, which approximates the uncertainty in the predictions. This sample selection process minimizes redundant simulations, accelerates model convergence, and significantly reduces the simulation overhead required for statistical characterization, thereby enhancing both prediction accuracy and efficiency.

III. Experimental Results and Discussion

The proposed BLAST was implemented in PyTorch and validated under TSMC 12nm process. Statistical setup and hold constraint characterization results by commercial tool SiliconSmart were taken as golden and used for model training and test under 135 PVT corners with 3 process corners, 9 voltages from 0.6V to 1.0V, and 5 temperatures from 0°C to 100°C. To demonstrate the generalizability of BLAST, the validation experiments were performed on various flip-flop structures [5, 6, 7] besides traditional MSFF. The logic structure reduction flip-flop (LRFF) [5] incorporates a logic structure reduction technique to SR-latch-based flip-flop. The sense-amplifier-based flip-flop with transition completion detection (SAFF-TCD) [6] optimizes the data input circuit with a sense amplifier-based circuit. The self-timed pulsed latch (STPL) [7] optimizes the clock input circuit with a pulse generation circuit.

TABLE I Prediction accuracy comparison. The unit of WCAE is *ps*.

flip-flop	RR [2]		FFNN [3]		BLAST	
	WCAE	rRMSE	WCAE	rRMSE	WCAE	rRMSE
MSFF	3.84	4.78%	2.27	3.02%	0.77	1.51%
LRFF [5]	4.11	5.27%	2.62	4.40%	0.74	2.03%
SAFF-TCD [6]	4.95	6.50%	3.07	5.88%	0.94	2.46%
STPL [7]	4.91	6.67%	3.69	5.26%	0.83	2.85%
Ave.	4.45	5.80%	2.91	4.64%	0.82	2.21%
	5.4×	2.6×	3.5×	2.1×	1.0×	1.0×

The proposed BLAST achieves a significant runtime speedup for constraint characterization compared to the commercial tool, as demonstrated in Fig. 3. During the training stage of BLAST, 0.8 hours were spent on extracting the specific feature by simulation, while 20.4 hours were consumed by traditional binary search iterations for sample labeling. In the inference stage, 3.8 hours were used to extract the specific feature. Compared to the simulation time for feature extraction and labeling, both the training and inference time of the MMoE model are negligible. With the incorporated active learning, the most informative samples were selected so as to maximize training efficiency. It can be found in Fig. 3 that with the proposed BLAST, 76.9% runtime is saved for statistical constraint characterization compared to the commercial tool, where in inference stage, the laborious bisection analysis is replaced by efficient simulation to extract the data path delay and clock path delay as specific feature.

The precision of the proposed BLAST is compared with the commercial tool as well as prior learning-based characterization methods including ridge regression (RR) [2] and feedforward neural network (FFNN) [3] in terms of the worst-case absolute error (WCAE) and the relative root mean square error (rRMSE) to represent the maximum and average prediction error respectively. As shown in Table I, the proposed BLAST demonstrates excellent accuracy across all validated PVT corners with a WCAE of 0.82 *ps* and a rRMSE of 2.21% for all cells [5, 6, 7]. Compared to the competitive models of RR [2] and FFNN [3], BLAST shows substantial improvements in prediction accuracy with the decrease of WCAE by 5.4× and 3.5× and rRMSE by 2.6× and 2.1×, respectively.

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